

Technical Document #187: Deep Learning - Comprehensive Overview

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Transformer architecture relies entirely on self-attention mechanisms, dispensing with recurrence and convolutions. It has become the foundation for modern NLP models.

Generative adversarial networks (GANs) consist of two neural networks competing against each other. The generator creates synthetic data while the discriminator evaluates authenticity.

Convolutional neural networks (CNNs) are designed to process grid-like data such as images. They use convolutional layers to automatically learn spatial hierarchies of features.

Batch normalization normalizes the inputs of each layer, reducing internal covariate shift. It allows higher learning rates and acts as a regularizer.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Key Takeaways:

- Recurrent neural networks (RNNs) are designed to process sequential data.
- Long short-term memory (LSTM) networks are a type of RNN that can learn long-term dependencies.
- Residual connections enable training of very deep neural networks by providing shortcuts for gradient flow.